Deep Learning Project

Multi-Modal Classification and Analysis of Images and Text using Deep Learning

Project Objectives

Students will develop a multi-modal pipeline to process image and text data using advanced deep learning techniques:

1. **Classify images** using a Convolutional Neural Network (CNN).
2. **Analyze textual descriptions** associated with images using word embeddings and Recurrent Neural Networks (RNNs).
3. **Fuse image and text information** to achieve optimal performance with a multi-modal approach.

In addition to implementation, students are required to **theoretically and mathematically describe the optimal model** for this task, explaining each component and architectural choice to maximize accuracy.

Dataset Choices

Students can choose from the following four datasets depending on the desired project complexity and data characteristics:

1. **MS COCO (Microsoft Common Objects in Context)**
    - **Description**: MS COCO is one of the most widely used datasets in computer vision, with over 120,000 labeled images and multiple captions per image. It offers diverse objects and contexts.
    - **Application**: Students can use CNNs for object classification and RNNs to analyze textual descriptions. This dataset is ideal for image-text association tasks.
    - **Link**: MS COCO Dataset
2. **Flickr8k/Flickr30k**
    - **Description**: These datasets contain 8,000 and 30,000 images respectively, with several captions per image. The captions are generally short and specific, simplifying text preprocessing.
    - **Application**: Flickr8k is a lighter option for beginners, while Flickr30k is suitable for medium-sized projects. Students can experiment with caption generation and image classification.
    - **Link**: Flickr8k Dataset and Flickr30k Dataset
3. **Fashion-Gen**
    - **Description**: This dataset contains clothing images paired with textual descriptions of style and categories. It is well-suited for exploring relationships between visual features and detailed descriptions.
    - **Application**: The CNN can classify types of clothing, while the text descriptions can be analyzed for style classification. This dataset is excellent for understanding visual and textual comprehension.
    - **Link**: Fashion-Gen Dataset

4. **Visual Genome**
    - **Description**: Visual Genome provides images annotated with detailed descriptions, object relationships, and visual and semantic metadata. It is more complex and rich in contextual data.
    - **Application**: Visual Genome is recommended for advanced students who wish to explore relationships between objects and descriptions. It is an excellent choice for complex multi-modal fusion tasks.
    - **Link**: Visual Genome Dataset

Steps to follow :

1. **Image Processing with CNN**
    - **Preprocessing**: Resize and normalize images.
    - **CNN Model**: Use a pre-trained CNN model (such as ResNet, VGG, or EfficientNet) to extract visual features. The model's fully connected layers will be fine-tuned for dataset-specific classes.
    - **Fine-Tuning**: Adjust the upper layers of the CNN model to improve performance for the dataset's classes.

2. **Text Processing with Embeddings and RNN**
    - **Preprocessing**: Tokenize text descriptions and convert words into embedding vectors using Word2Vec, GloVe, or a contextual embedding model (like BERT).
    - **RNN Model**: Pass embedding vectors through an RNN (GRU or LSTM) to capture sequential and contextual dependencies in the text descriptions.
    - **Sequence Encoding**: The final layer of the RNN will represent the encoding of the text associated with each image.

3. **Multi-Modal Fusion**
    - **Concatenation of Encodings**: Visual features extracted by the CNN and text sequence encodings produced by the RNN will be fused by concatenating their output vectors.
    - **Fully Connected Layers**: Pass concatenated vectors through several fully connected layers to learn interactions between image and text data.
    - **Softmax for Final Classification**: The last layer will be a softmax layer that assigns the probability of each class for each image-text pair.

4. **Model Tuning and Optimization**
    - **Fine-Tuning of Pre-Trained Models**: Adjust parameters of the CNN and RNN models to achieve better performance. Test different CNN models (ResNet, EfficientNet) and RNNs (LSTM, GRU).
    - **Hyperparameter Tuning**: Use a tuning technique (such as Grid Search or Random Search) to optimize model hyperparameters, including learning rate, batch size, and the number of fully connected layers for fusion.

  - **Regularization**: Apply regularization techniques like dropout in the fully connected layers to prevent overfitting.
  - **Evaluation with Cross-Validation**: Evaluate model performance with cross-validation to ensure generalization.

5. Student-Proposed Dataset (Custom Topic + Dataset Choice)
Students may propose their own topic along with a dataset, especially for exploring emerging areas such as Generative AI or domain-specific multi-modal tasks.

Examples can include:

- fashion recommendation with synthetic captions,
- image–text sentiment analysis,
- cultural heritage images with expert descriptions,
- medical images with reports (if ethically compliant),
- or datasets enriched using GenAI (synthetic captions or augmented samples).

Theoretical and Mathematical Description of the Optimal Model

Students should also **describe the optimal model theoretically and mathematically**:

- **CNN and RNN**: Mathematically describe convolutions, pooling, activation functions, and the activation equations in an LSTM or GRU, explaining the gates (input, forget, and output) in an LSTM.
- **Loss Function and Optimization**: Explain cross-entropy loss and the optimizer's role in minimizing it.
- **Multi-Modal Fusion**: explain the concatenation process and explain the importance of a fully connected layer to capture correlations between images and texts.

Deliverables

1. **Theoretical and Mathematical Report**:
   - A complete description of the chosen optimal model, with theoretical and mathematical details of each component.
2. **Source Code**: Well-documented code to train the model and evaluate its performance.
3. **Presentation**:
   - A final presentation of the results with a discussion on model choices, optimization, and the impact of multi-modal fusion on performance.